Code-Layout, Readability and Reusability

Date	30 September 2026
Team id	SWTID-2026-2551
Project name	Fit Buddy – AI Fitness Plan Generator using Gemini Models
Maximum mark	5 marks

Code Layout Checklist

The checklist follows the same quality parameters in your uploaded template

S .No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	**Consistent Indentation**	Uniform indentation and spacing are used throughout the frontend and backend code.	Yes	Consistent formatting is maintained across React and Fast API files.
2	**Proper File Structure**	Files and folders are logically organized according to their functionality.	Yes	Components, pages, services, routes, models and configurations are separated into appropriate folders.
3	**Meaningful Variable Names**	Variables clearly reflect their purpose.	Yes	Names such as fitness Goal, activity Level, user Profile and generated Plan are used.
4	**Function / Method Names**	Functions are descriptively named according to their responsibilities.	Yes	Functions such as generate Fitness Plan(), create User() and authenticate User() are used.
5	**Code Comments**	Inline and block comments explain important or complex logic.	Yes	Comments are provided for Gemini integration, API logic

				and important processing steps.
6	**Modular Design**	Code is divided into reusable functions, components and modules.	Yes	React components, Fast API routes and Gemini service modules are separated.
7	**No Redundant Code**	Duplicate, unused or unnecessary code is avoided.	Yes	Common functionality is placed in reusable components and services.
8	**Error Handling**	Exceptions and API errors are handled appropriately.	Yes	Backend validation, API error handling and Gemini API failure handling are included.

Reusable Components / Modules

s.no	**Component / Module Name**	**Language / Technology**	**Where Reused**	**Reusability Level**
1	**Fitness Form**	React.js	Fitness profile and fitness plan generation pages	High
2	**Plan Card**	React.js	Workout, nutrition and recommendation sections	High
3	**API Service**	JavaScript / Axios	All frontend API requests	High
4	**Authentication Service**	Python / Fast API	Login, registration and protected APIs	High
5	**Gemini Service**	Python / Gemini API	AI fitness plan generation	High

Overall Code Quality Assessment

Aspect	**Rating (1–5)**	**Comments**

Code Layout & Structure	5	Frontend, backend, services, components and configuration files are logically organized.
Readability	5	Meaningful names, consistent formatting and clear function responsibilities improve readability.
Reusability	5	Common UI components, API services and AI services are designed for reuse.
Documentation / Comments	4	Important functionality and complex AI/API logic are documented with comments.
Overall Score	**5 / 5**	The Fit Buddy codebase follows a structured, readable and modular development approach.

This final section matches the **Overall Code Quality Assessment** format in your template